HACKATHON SUBMISSION

AEGIS

AI-Enhanced Guardian Interface System

Team details

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Mission

Bridging the gap between trauma and treatment through Augmented Reality delivering real-time, AI-powered medical guidance directly to the soldier at the point of injury.

THE PROBLEM

The Golden Hour Is Slipping Away

The critical gap in battlefield survival is the "Platinum Ten"—the first ten minutes following an injury where 90% of preventable combat deaths occur before a professional medic can arrive. Under the extreme stress of active fire, non-medical soldiers often experience cognitive collapse and task saturation, leading to fatal procedural errors in basic life-saving steps like tourniquet application or airway management. Furthermore, performing these interventions in total darkness presents a tactical catch-22: using white light reveals the squad's position to the enemy, while attempting treatment without it leads to high failure rates. Currently, there is no hands-free, real-time system capable of bridging this expertise scarcity by providing a soldier with immediate, specialist-level surgical guidance at the exact moment of injury

🩸 The Critical Gap

90% of preventable combat deaths occur at the Point of Injury — before a medic ever reaches the scene.

🧠 The Cognitive Load

Extreme stress degrades decision-making. Soldiers panic under pressure, failing to correctly apply tourniquets, chest seals, or wound packing.

🌑 The Visibility Crisis

Standard medical intervention is nearly impossible in zero-light environments — using white light compromises the soldier's tactical position.

THE SOLUTION

Sentinel-Medic AR: Eyes, Brain, and Guidance — On the Helmet

Computer Vision Triage

An AI-powered HUD that analyzes wounds in real-time, classifying injury type — arterial bleed, fracture, penetrating trauma — instantly and automatically.

Dynamic AR Guidance

No more static field manuals. The soldier sees step-by-step visual overlays — arrows, annotations, and instructions — projected directly onto their field of vision.

Digital Night Vision

An IR-modified sensor enables full medical intervention in total darkness — zero white light, zero position compromise, full situational stealth.

ARCHITECTURE

The Pipeline: From Wound to Guidance in Milliseconds

The entire loop — from camera capture to AR overlay — runs on low-latency edge hardware, requiring no cloud uplink in the field. Every second saved is a life closer to being saved.

TECH STACK

Built on Battle-Ready Technology

⚙️ Hardware Layer

Vision Node

ESP32-CAM-MB — compact, low-power edge camera

Sensors

NEO-6M GPS · MAX30102 Vitals · INMP441 Audio

Optics

Stereoscopic VR chassis + high-resolution smartphone display

🧠 Software & AI Layer

- **Vision:** OpenCV & YOLOv8-tiny
- **Orchestration:** Python — FastAPI / Flask
- **LLM / RAG:** Gemini 1.5 Flash (Vertex AI) + MongoDB Atlas Vector DB
- **Infrastructure:** Google Cloud Platform — Cloud Functions